

PROJECT DESIGN PHASE II

TECHNOLOGY STACK

(ARCHITECTURE & STACK)

Date	28.10.2025
Team ID	NM2025TMID04506
Project Name	Laptop Request Catalog item
Maximum Marks	4 Marks

TECHNICAL ARCHITECTURE:

The technical architecture for the "Laptop Request" item is contained entirely within the ServiceNow Cloud (SaaS) platform.

Processes (Application Logic): The primary logic is handled by ServiceNow Flow Designer. This flow manages the state of the request, orchestrates the approval process, and triggers the creation of the fulfillment task.

Infrastructure Demarcation (Local/Cloud): The User interacts with the system via their local web browser, which accesses the Service Portal (in the cloud). All application logic and data processing occurs within the ServiceNow Cloud.

External Interfaces: The system interfaces with an Email Notification service (part of the platform) to send approval and status update emails to the Manager and User.

Data Storage: All data, including the submitted Request Item (RITM), the approval record, and the IT fulfillment task (SC_TASK), is stored in the ServiceNow Cloud Database (e.g., in the sc_req_item and sc_task tables).

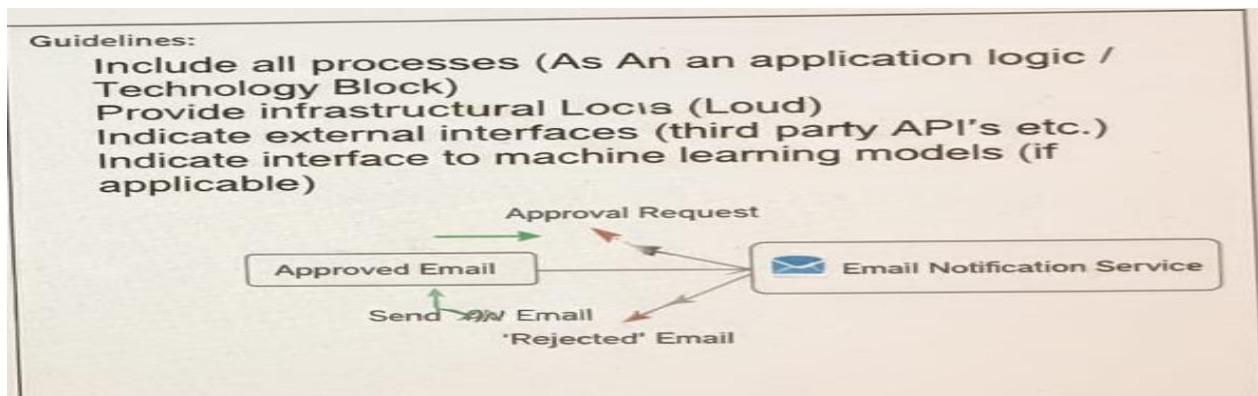
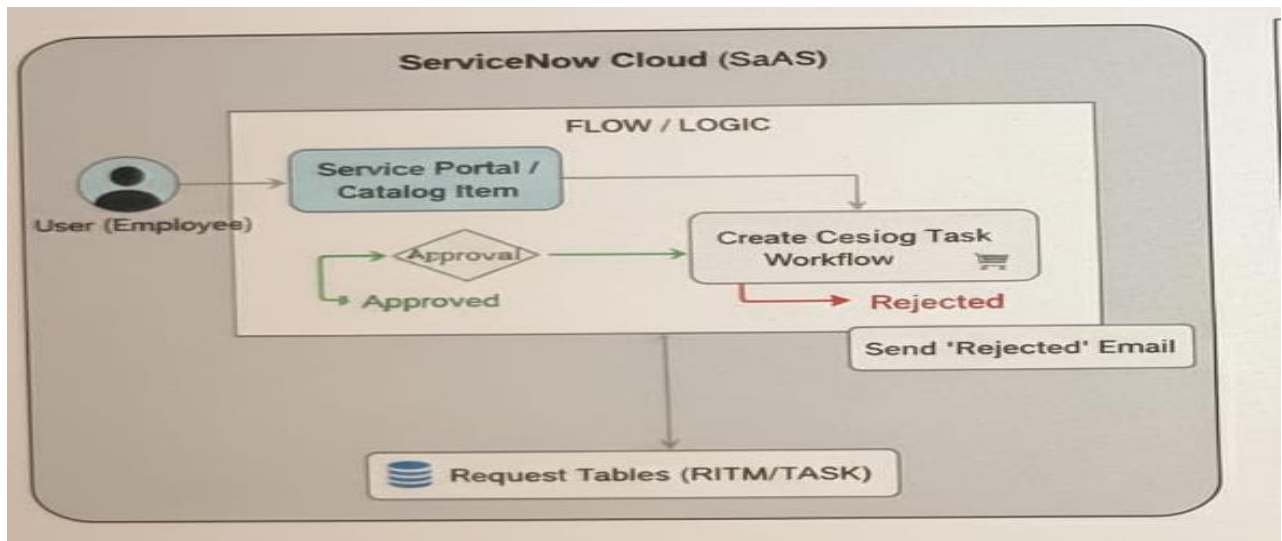


TABLE 1: COMPONENTS & TECHNOLOGY

S.No	COMPONENT	DESCRIPTION	TECHNOLOGY
1.	User Interface	Employee interacts with the Service Catalog to submit the request. Manager interacts with email. IT Fulfiller uses the backend platform.	ServiceNow Service Portal, Email Notifications, ServiceNow Platform UI
2.	Application Logic-1	Presents the form, variables (e.g., laptop choices), and auto-populates user data (Manager, Location).	Catalog Item, Variables, Catalog UI Policies

3.	Application Logic-2	Orchestrates the entire process: sends approvals, creates tasks, and updates the request status.	ServiceNow Flow Designer
4.	Application Logic-3	Sends event-based emails to the user (Submitted, Approved, Rejected, Completed).	ServiceNow Notifications
5.	Database	Stores the submitted request (RITM), the manager's approval record, and the fulfillment task (SC_TASK).	Request Tables (sc_req_item), Task Tables (sc_task), Approval Tables
6.	Cloud Database	Managed by the ServiceNow cloud backend.	ServiceNow Cloud Database
7.	External API	Not applicable for this project.	N/A
8.	Machine Learning	Model Not applicable for this project.	N/A
9.	Infrastructure	Hosted and managed on the ServiceNow SaaS platform.	ServiceNow Cloud (SaaS)

TABLE 2: APPLICATION CHARACTERISTICS

S.No	CHARACTERISTICS	DESCRIPTION	TECHNOLOGY
1.	Open-Source Frameworks	Not applicable (ServiceNow is a proprietary platform).	-

2.	Security Implementations	Role-based access (User Criteria) limits who can order the item. ACLs protect the backend task and request records.	User Criteria, Access Control Lists (ACLs)
3.	Scalable Architecture	SaaS-based, horizontally scalable via the ServiceNow Cloud infrastructure.	ServiceNow Cloud Architecture
4.	Availability	Highly available via ServiceNow's load-balanced, mirrored data centers.	Load-balanced ServiceNow Instances
5.	Performance	Optimized by using Flow Designer (asynchronous processing) and efficient, auto-populating scripts on the form.	Flow Designer, Catalog Client Scripts